

# METABOLISM, ENZYME, AND ATP

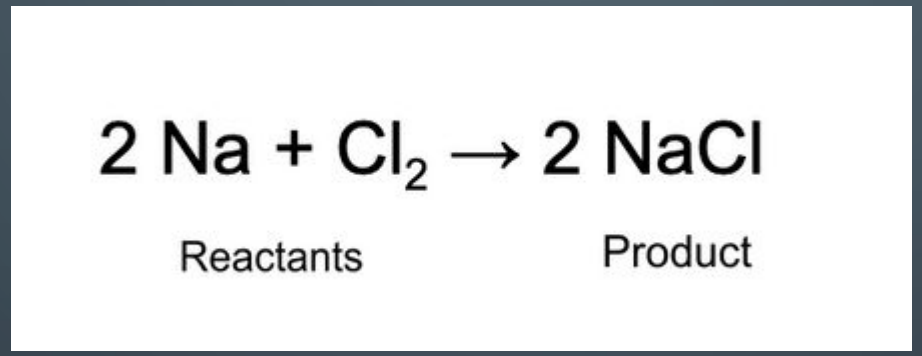
# METABOLISM VS. CATABOLISM

- **Metabolism**= a sum of all chemical reactions that occur within an organism
- **Catabolism**= the process of breaking down complex molecules into simpler compounds



# CHEMICAL REACTIONS

- Process that changes one set of chemicals into another set of chemicals

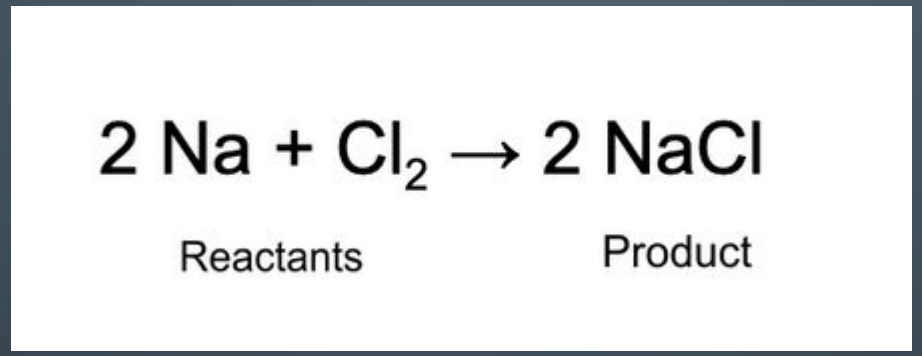



- Law of Conservation of Matter: matter cannot be created or destroyed

What happens to atoms during a chemical reaction?

# CHEMICAL REACTIONS

- Process that changes one set of chemicals into another set of chemicals



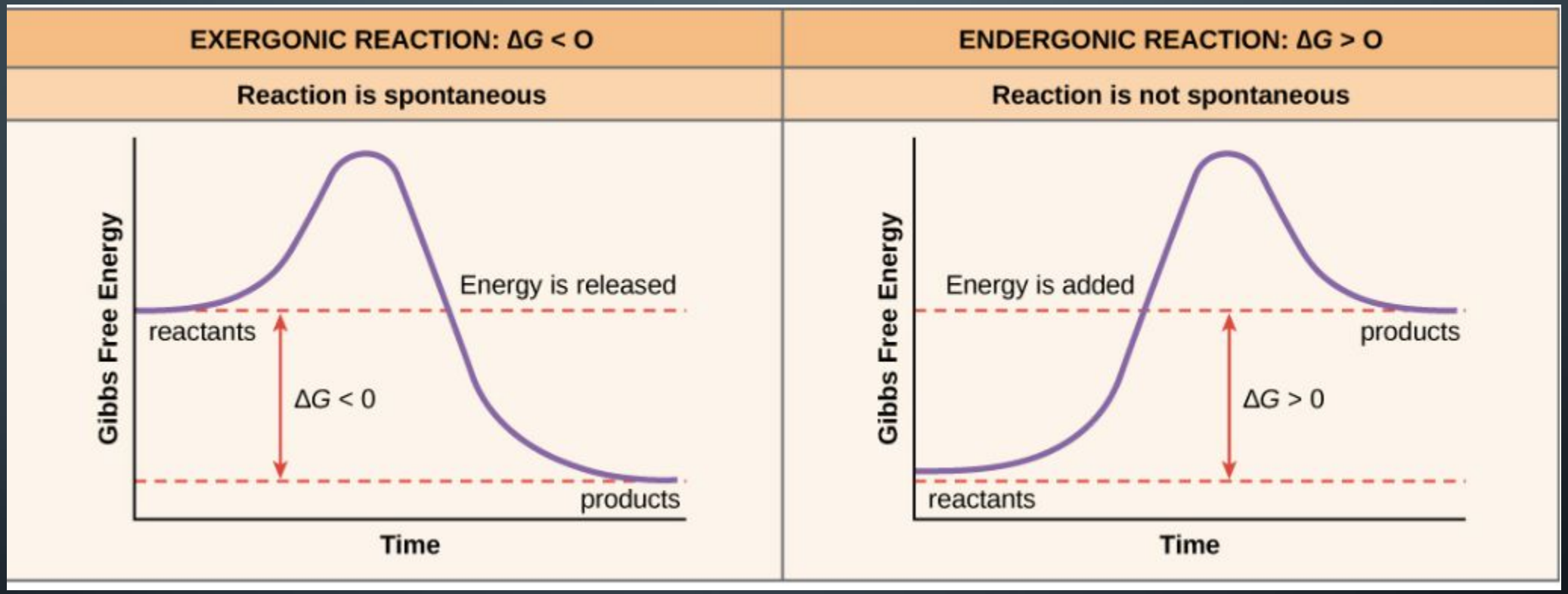

- Law of Conservation of Matter: matter cannot be created or destroyed

What happens to atoms during a chemical reaction? Atoms are rearranged

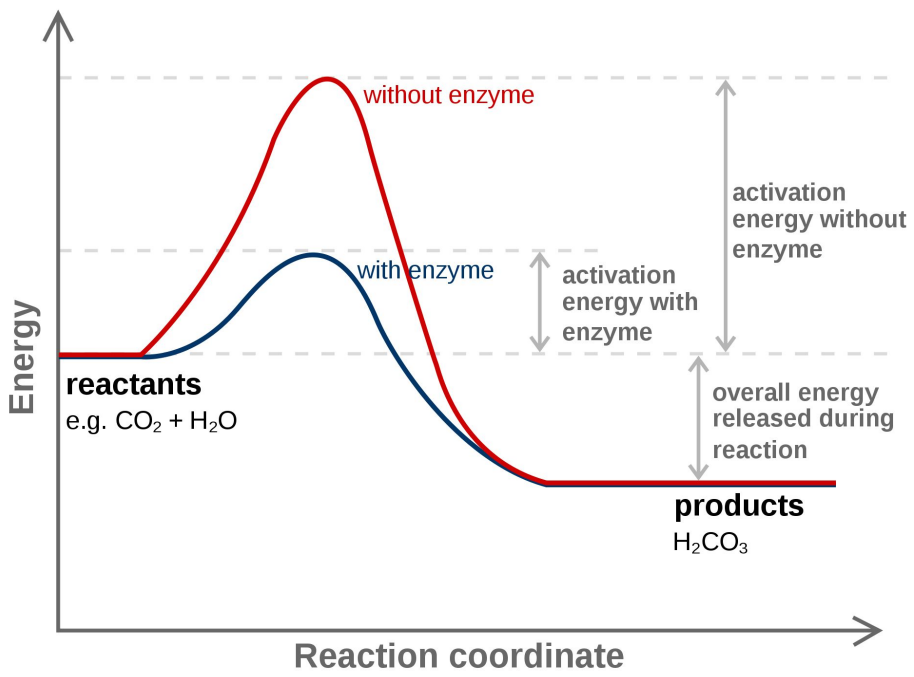
# Energy in Reactions

- Chemical reactions store or release energy depending on the type of reaction
- Exergonic Reaction: a release of energy occurs → the reactants will have more energy than the products
- Endergonic Reaction: require the input of energy to occur → the products will have more energy than the reactants.

# Energy Graphs

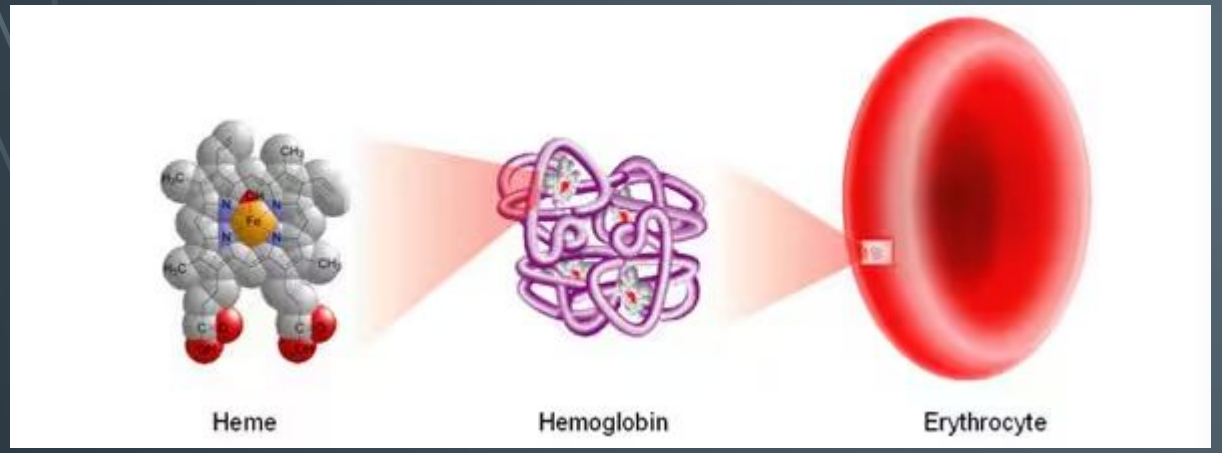


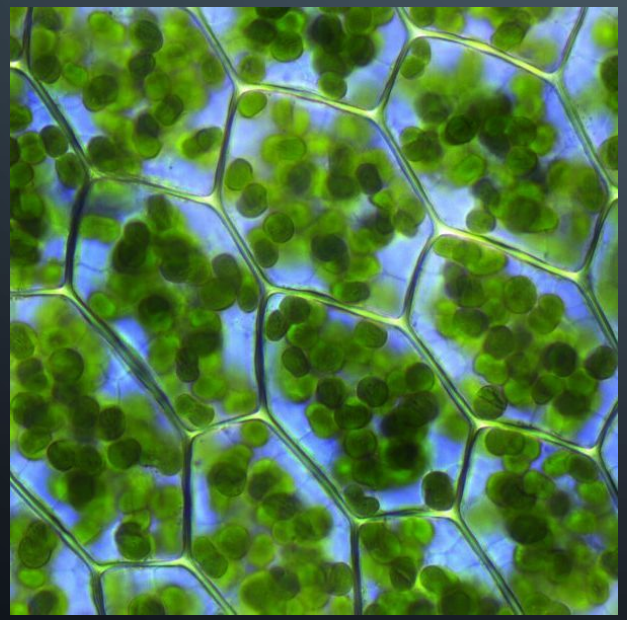

Which reaction would be anabolic and which reaction would be catabolic?




# ENZYMES

- Some chemical reactions are too slow or have too high an activation energy
- **Activation energy**: energy needed to start a reaction
- **Catalyst**: substance that speeds up the rate of the reaction by lowering activation energy required
- **Enzymes**: proteins that are catalysts- speed up chemical reactions by lowering activation energy

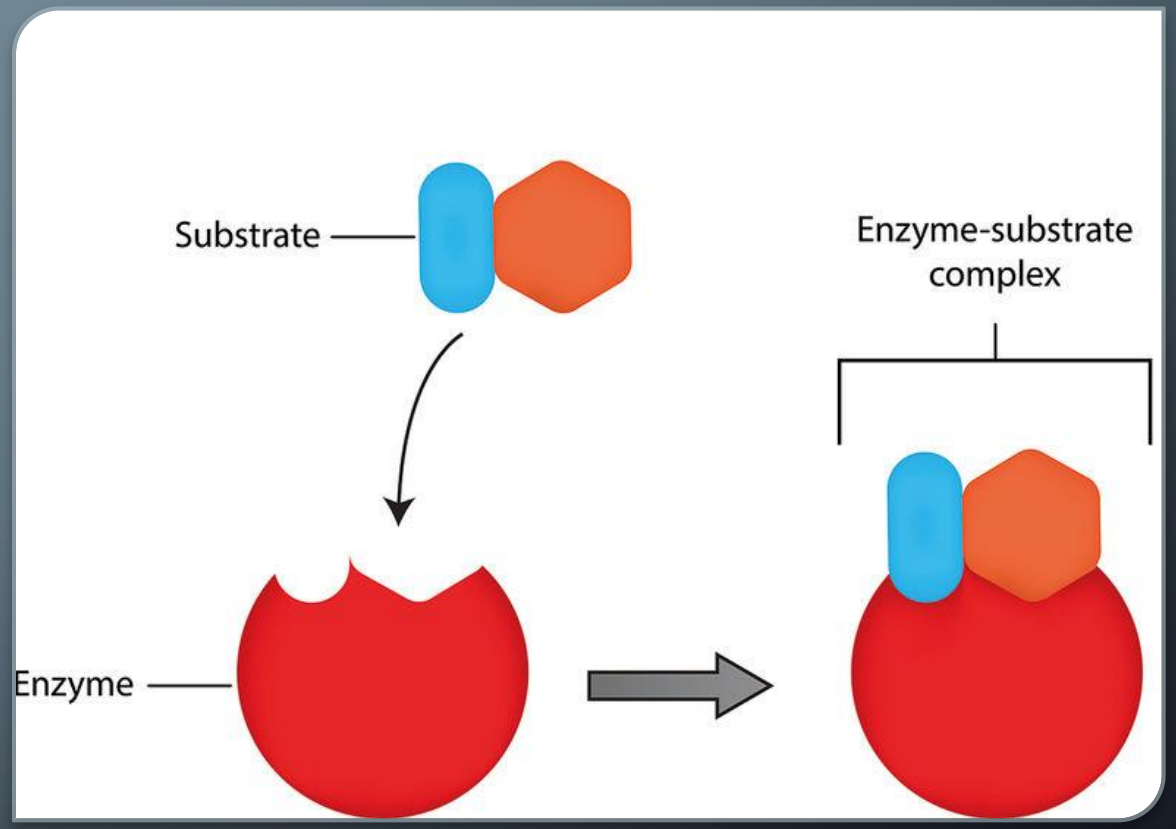


A study done by the UNC school of Medicine concluded that the synthesis of hemoglobin and chlorophyll would take approximately **2.3 BILLION years** without the presence of an enzyme.

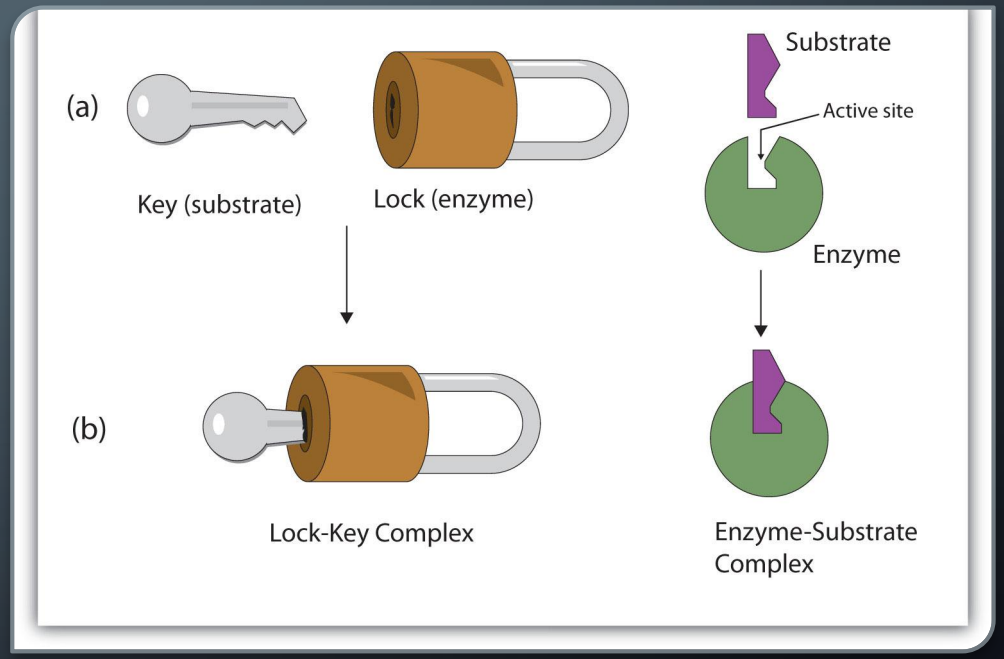
# ENZYME SUBSTRATE COMPLEX

- Substrate: reactants
- Active site: place where the substrates bind to the enzyme
- Enzyme + bonded substrates= Enzyme substrate complex

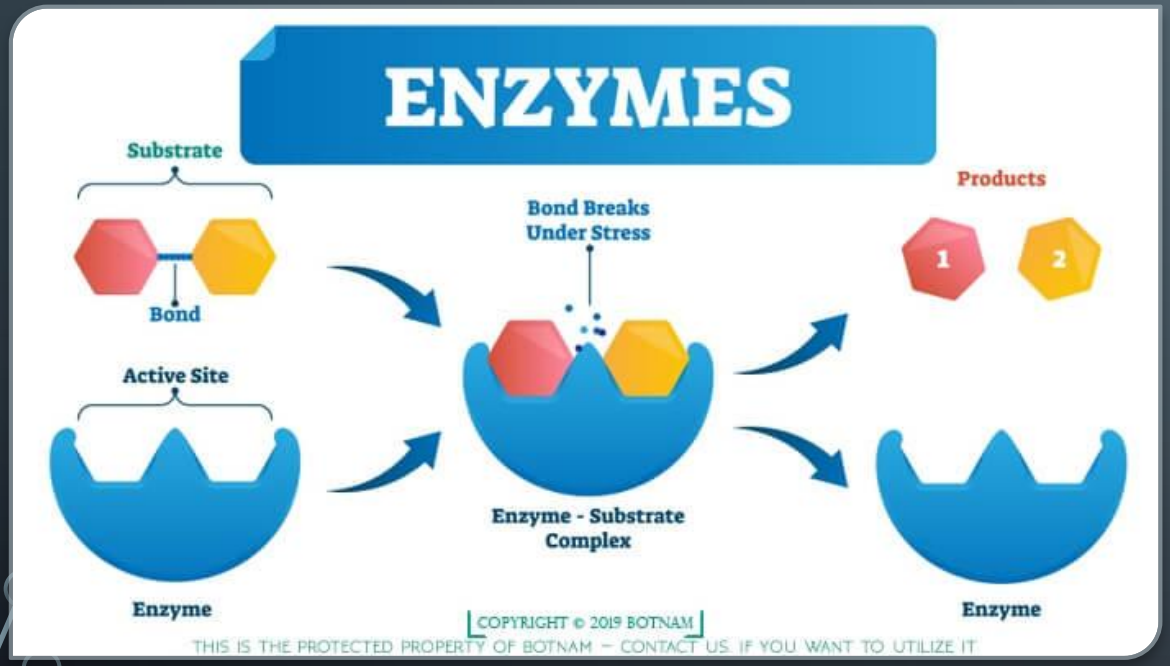


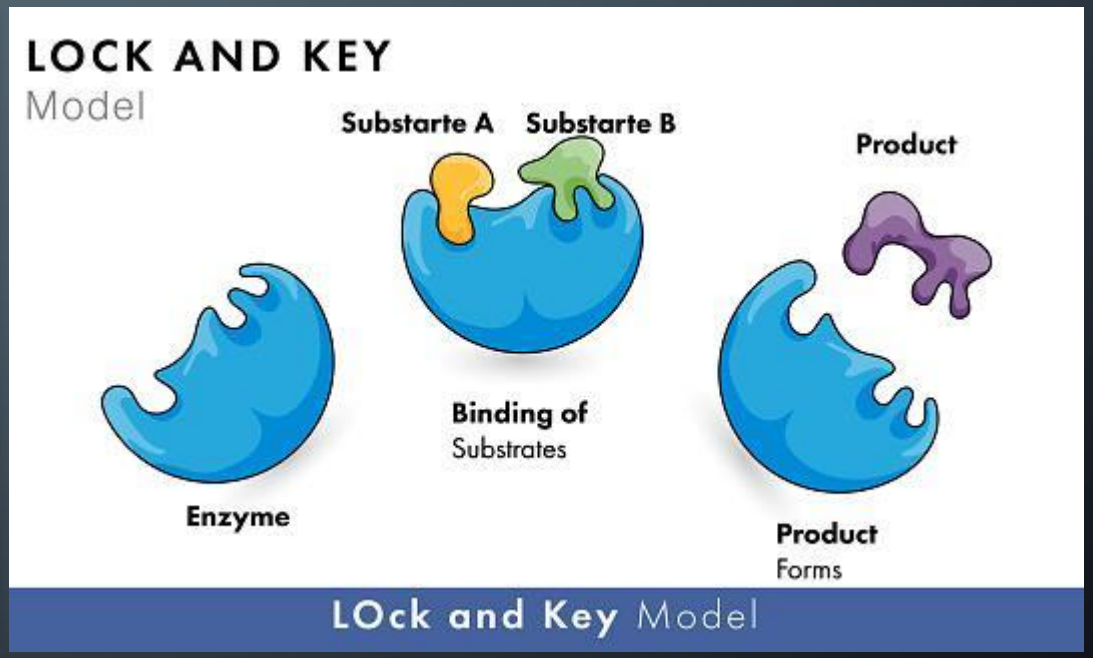
# LOCK AND KEY MECHANISM

- Enzymes can only bind to certain substrates based on their shape
- When the reaction is over, the enzyme is left as it was
  - Enzymes are NOT permanently changed or used up



# ENZYMES CAN CATALYZE CATABOLIC OR ANABOLIC REACTIONS



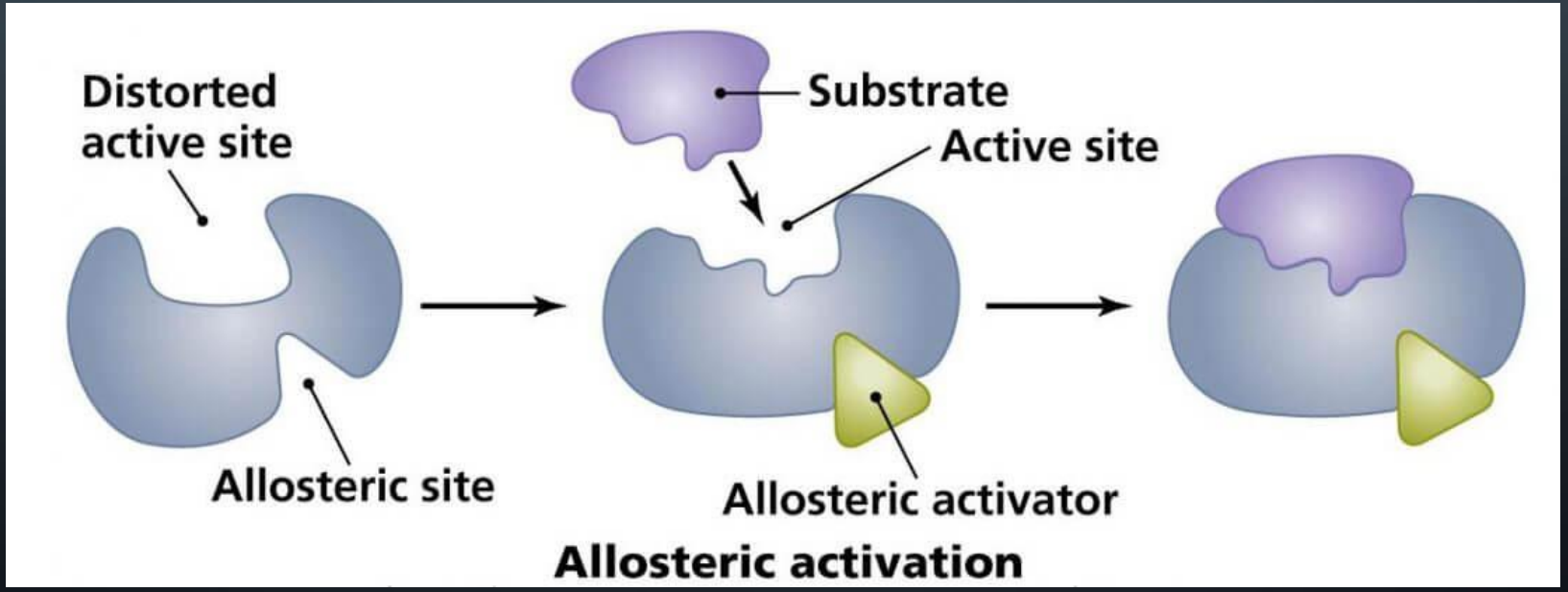




Catabolic 
-Breakdown
-Release energy

Anabolic 
-Build up
-Store energy

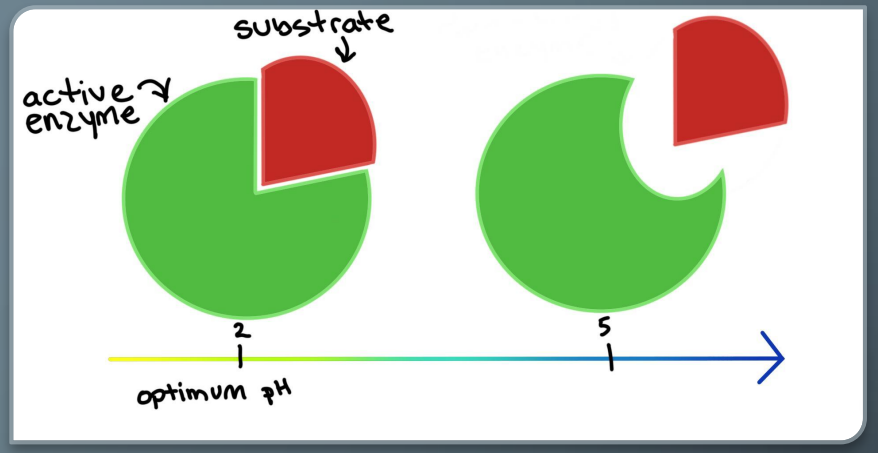
# ALLOSTERIC ACTIVATION

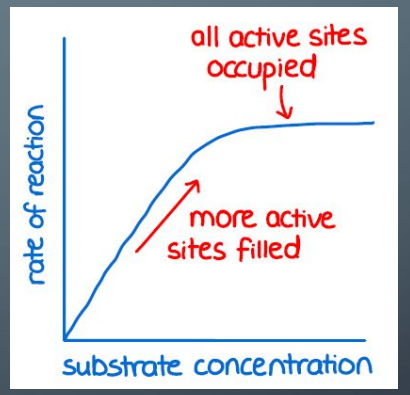
- Allosteric enzymes will have an allosteric (secondary) site necessary for binding
- Effector molecules bind to this site and modify the shape of the enzyme to allow the substrate to bind

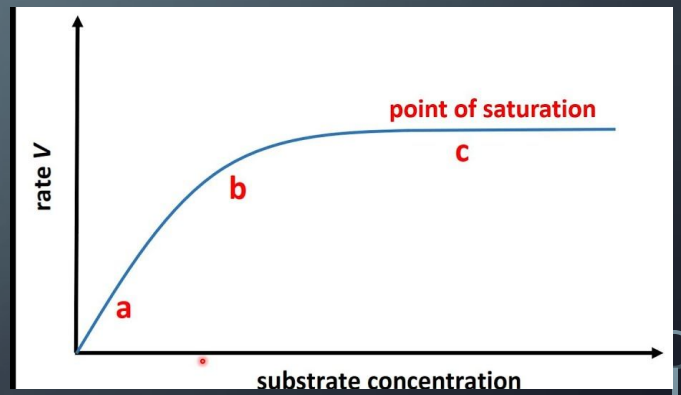


# CHANGES TO ENZYME ACTIVITY

- Some factors may affect the enzyme's ability to function by changing the **shape** of the enzyme
- Enzymes are sensitive to changes in temperature and pH
- Too much change results in **denaturation**- destruction of protein structure
- Amount of substrate available will also affect how active an enzyme is* →



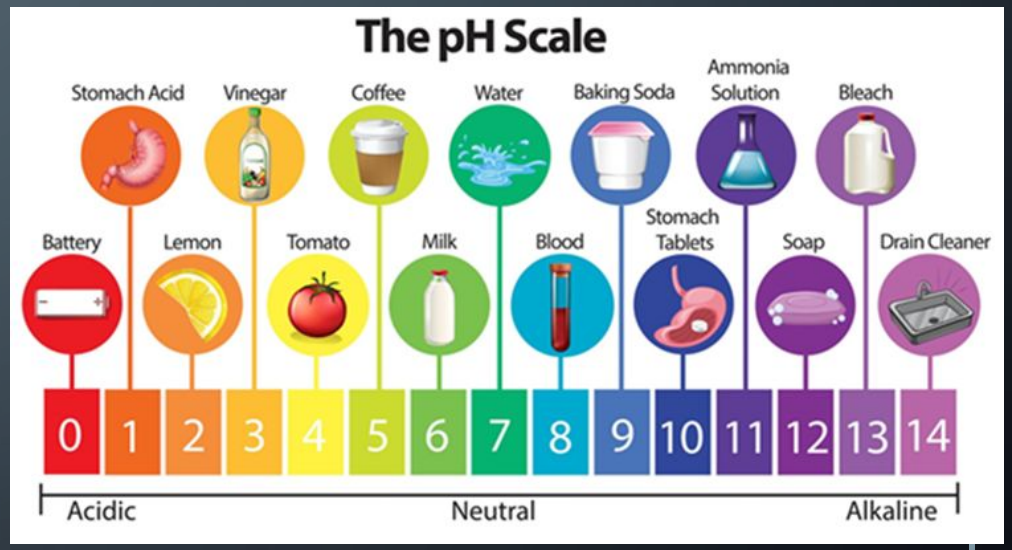




*Increasing substrate concentration increases the rate of reaction to a point. At some point the enzyme is considered "**saturated**" with substrate and the reaction rate will plateau.

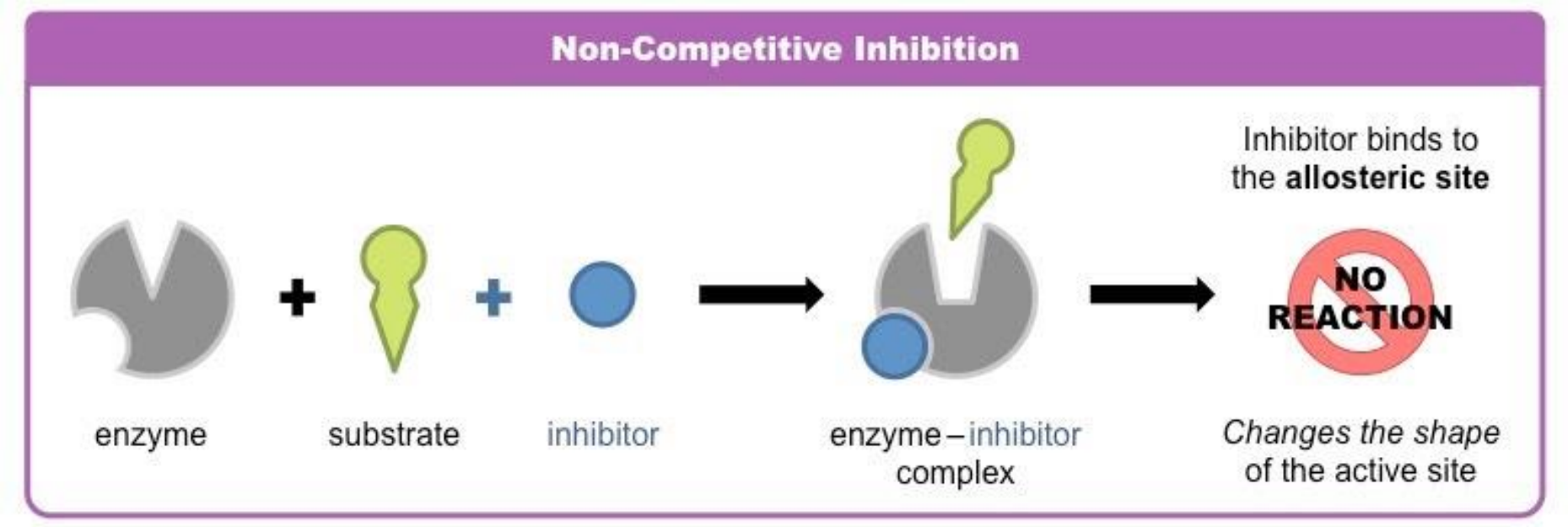
# The pH scale

- Used to rank solutions in terms of acidity or alkalinity (how basic it is)
- Measures concentration of H+ ions
- MOST solutions fall on this scale but it is possible for solutions to be less than zero or more than 14
- Most cells exist within a pH of 7 and exposure to more acidic or basic conditions can cause denaturation of proteins



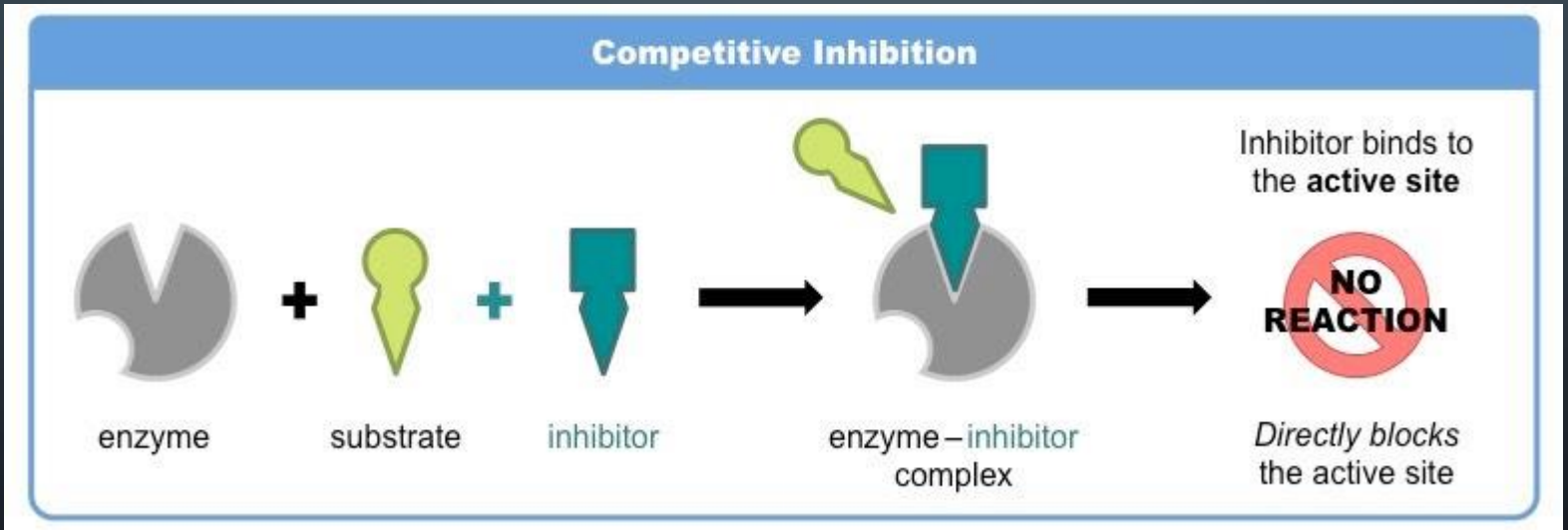
# NON-COMPETITIVE INHIBITION

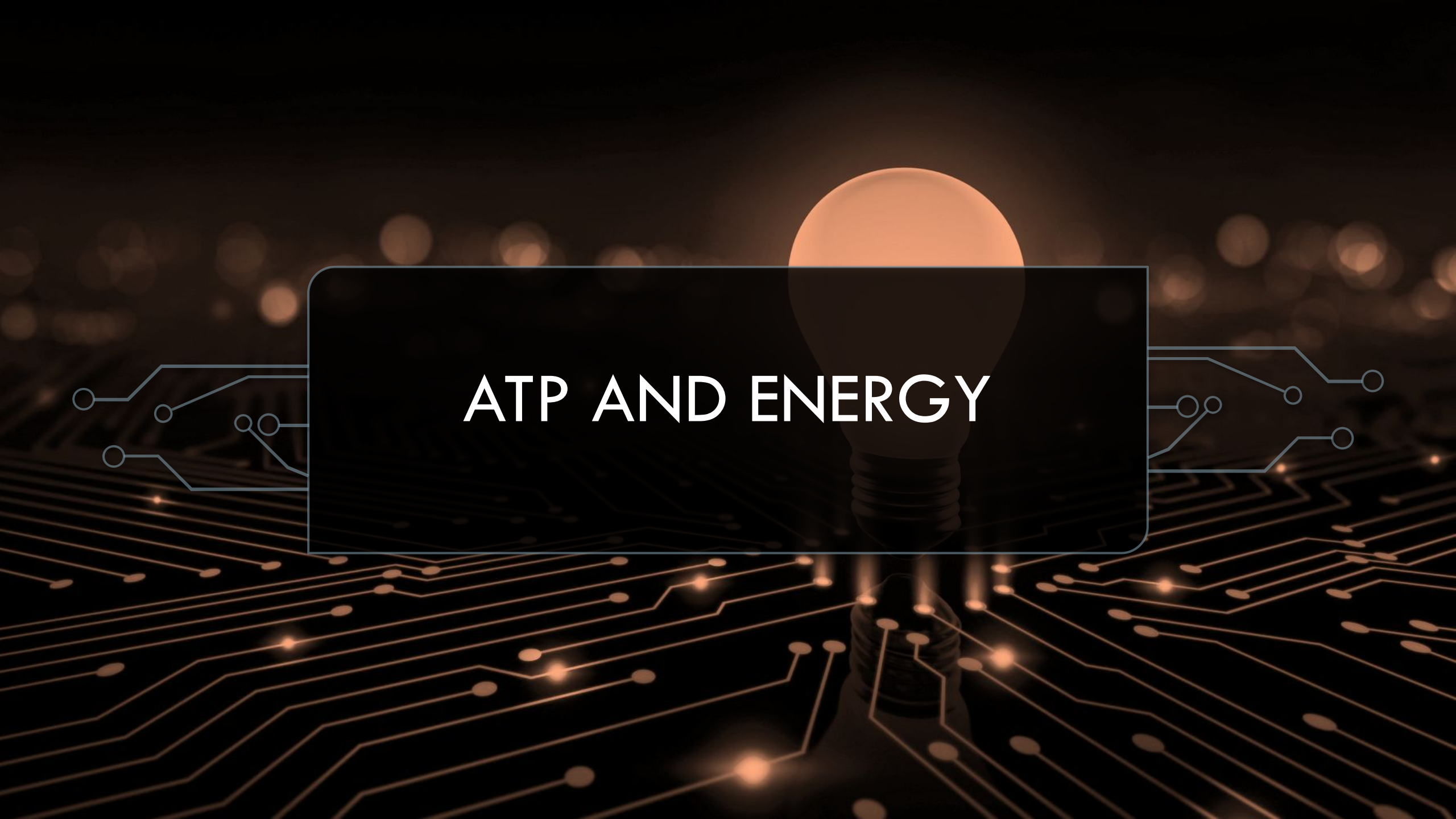
- An inhibitor molecule changes the shape of the enzyme by binding to the allosteric site
- Substrate is unable to  bind due to shape change



# COMPETITIVE INHIBITION

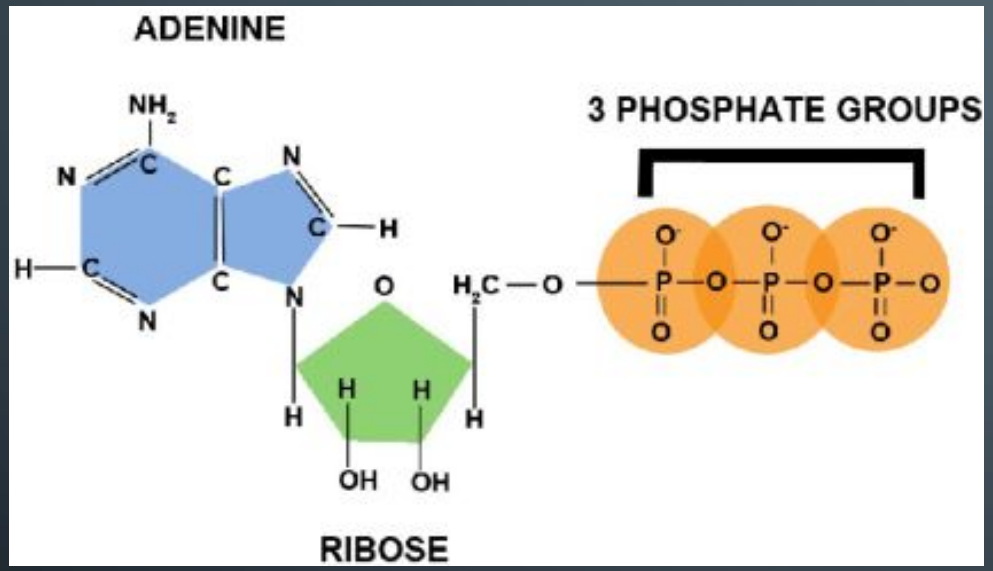
- An inhibitor molecule binds to the active site and prevents the substrate from binding at all → directly blocking the active site





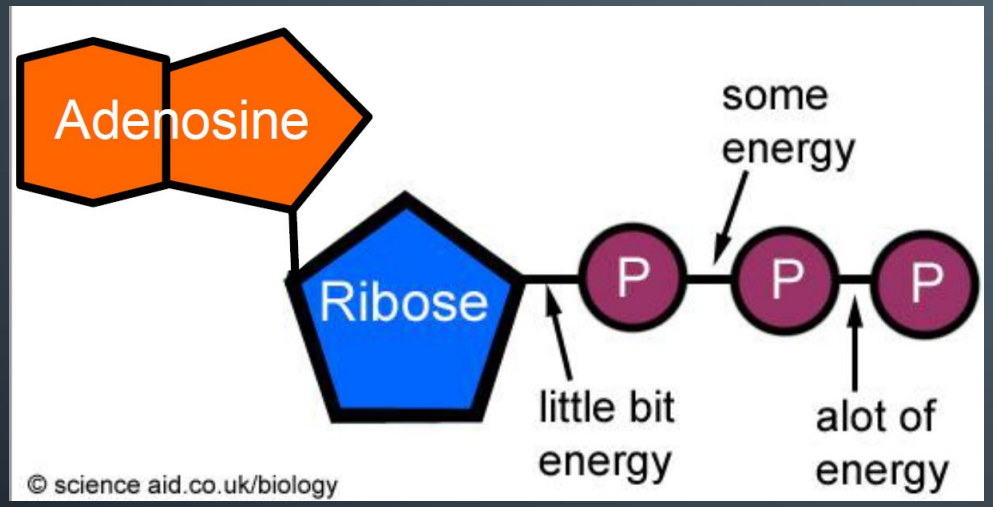
# ATP AND ENERGY

# ATP



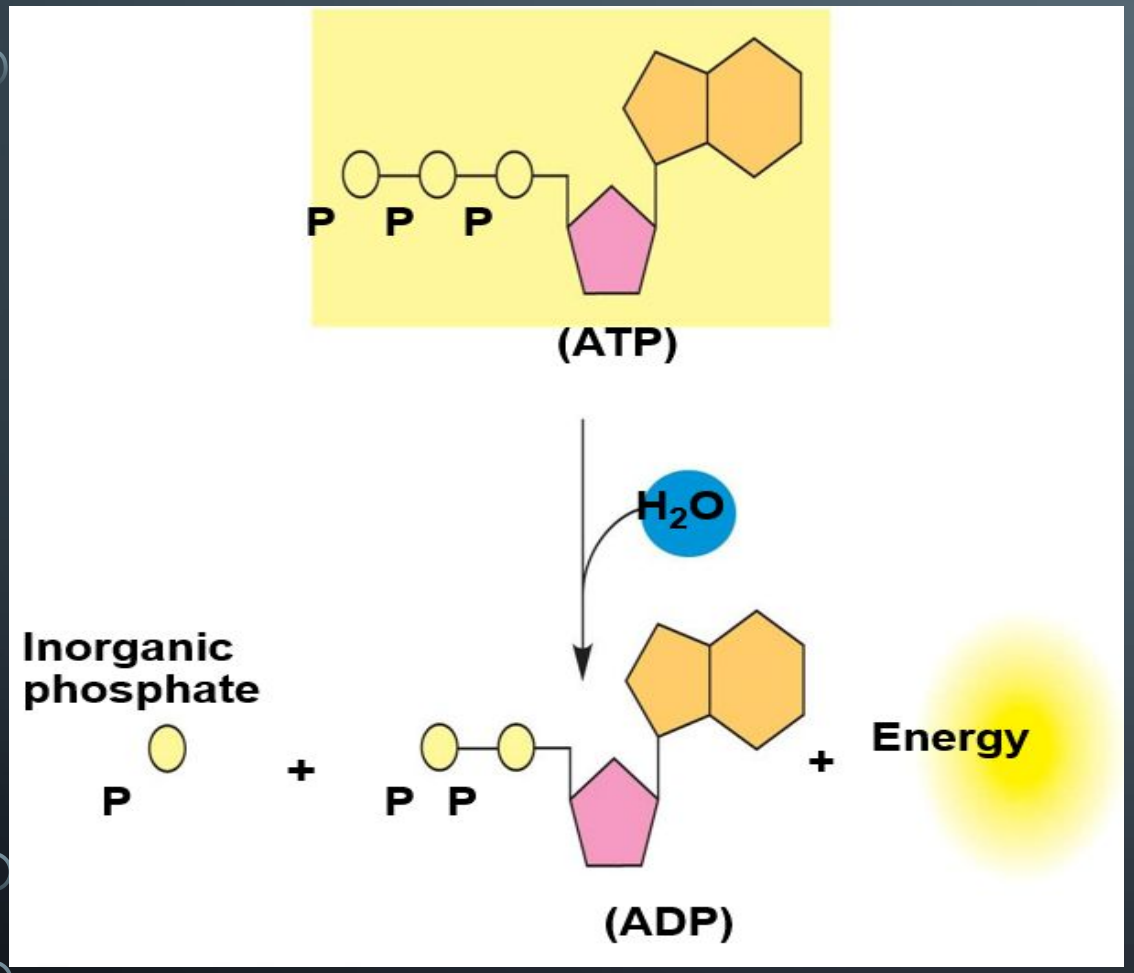

- Adenosine Triphosphate= cell's energy shuttle
- Composed of:
  - A ribose sugar
  - Adenine
  - 3 phosphate groups

# ENERGY FOR THE CELL



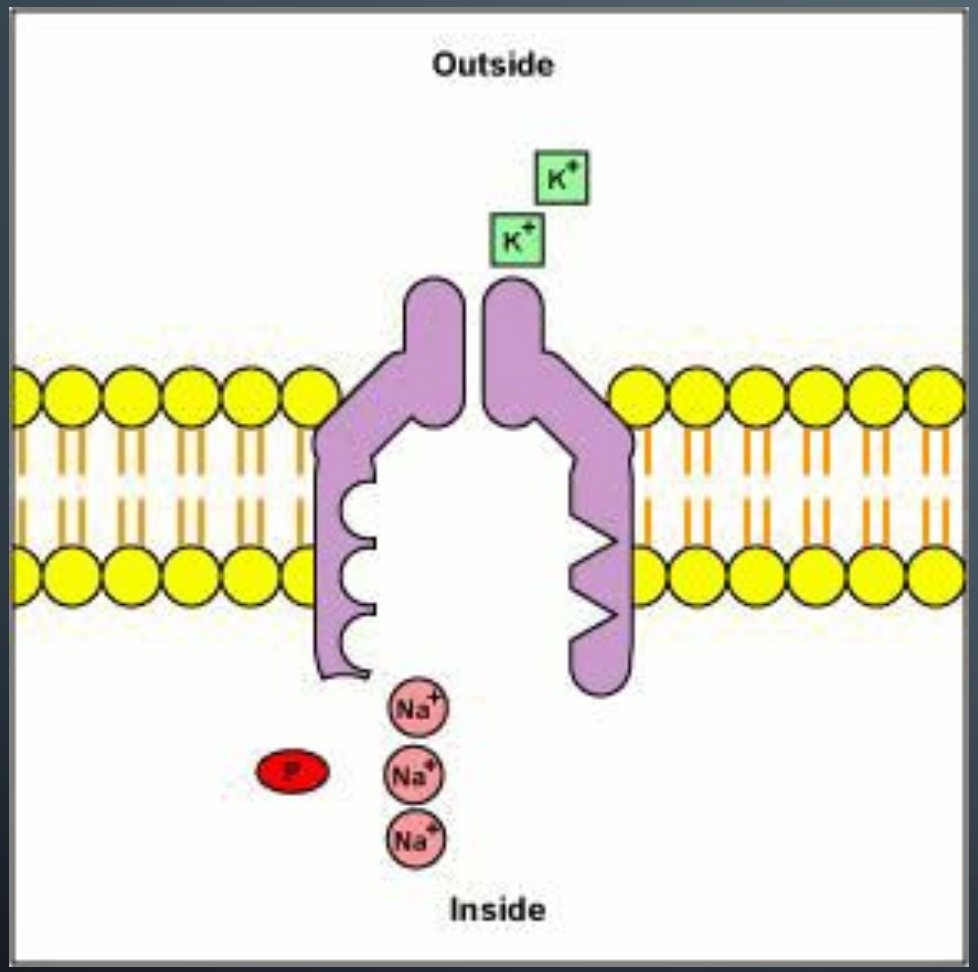

- ATP is chemical fuel that powers the activities of the cell.
- Energy is stored in the phosphate bonds

**Most energy in the bond between the second and third phosphate



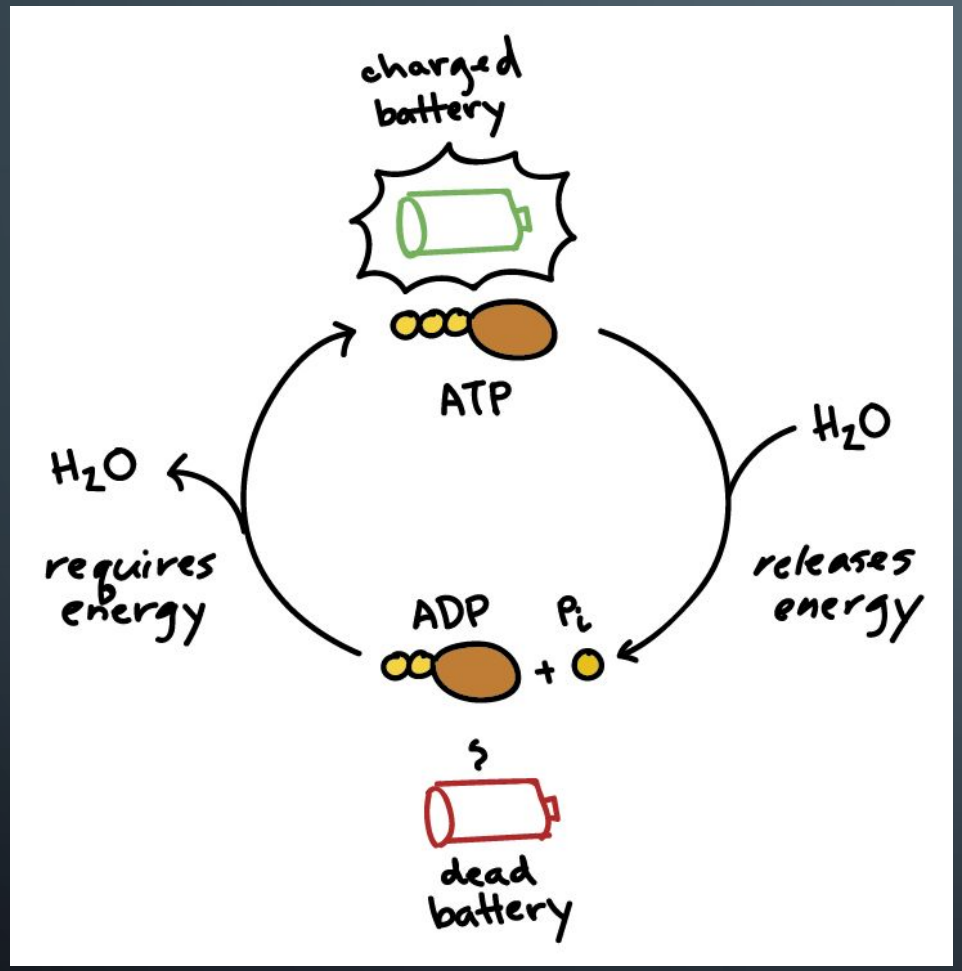

# ATP HYDROLYSIS

- Bonds between the last two phosphate groups in the tail of ATP are broken
- Energy is released as a result



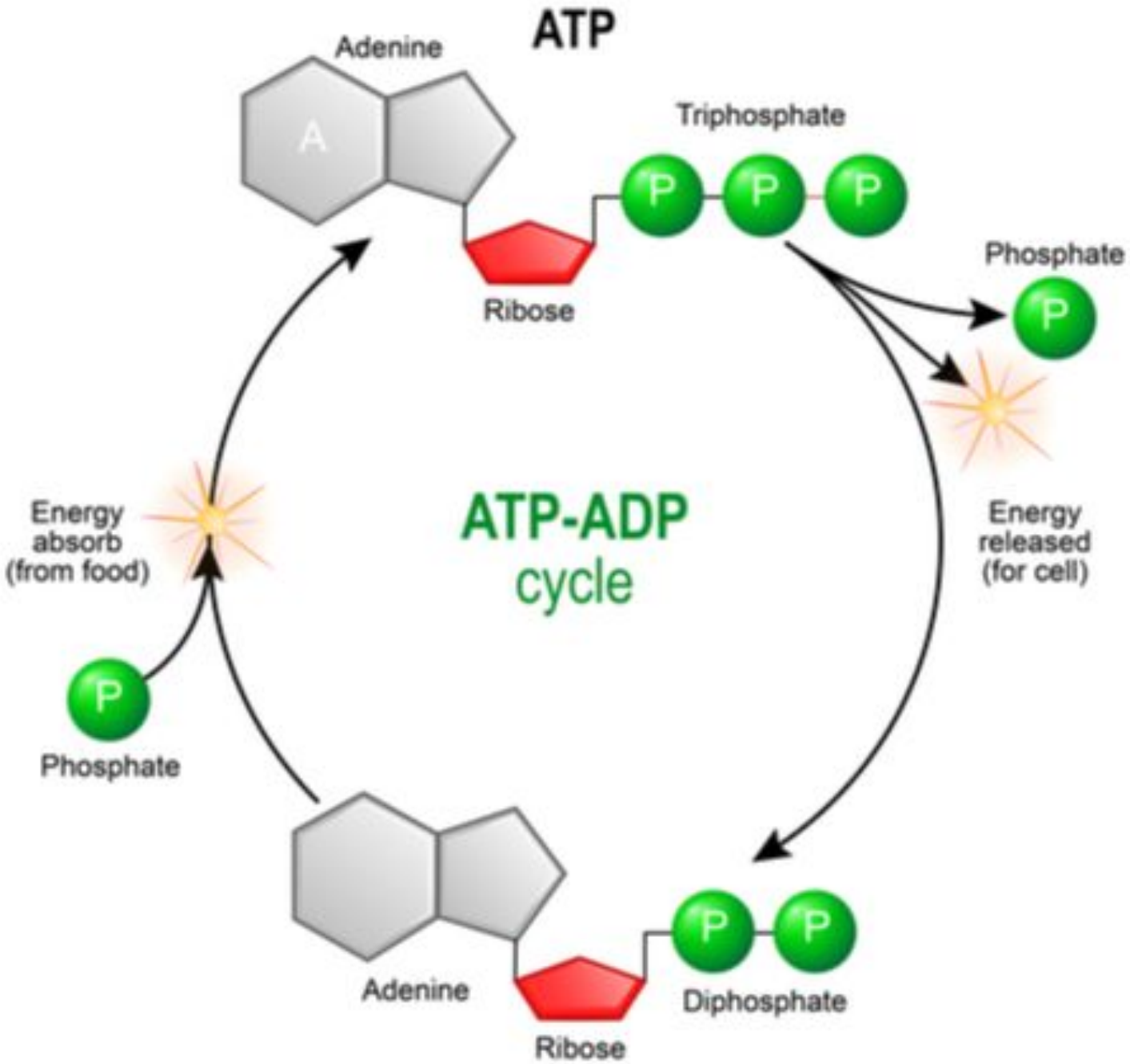

# WHAT HAPPENS TO THE PHOSPHATE GROUP?

- ATP hydrolysis helps to drive reactions by phosphorylation:
  - Transferring a phosphate group to some other molecule such as a reactant
- The recipient molecule is now phosphorylated  receives energy




# ATP ADP CYCLE

- ATP is renewable and regenerated by the addition of a phosphate group to adenosine diphosphate
- The energy to phosphorylate ADP comes from catabolic reactions in the cell
- Chemical energy stored in ATP is what powers most cellular work




# ATP-ADP CYCLE



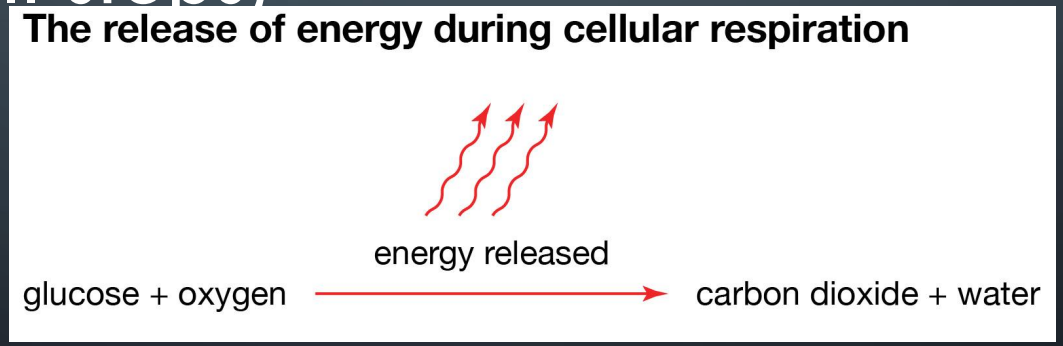


# HARVESTING STORED ENERGY

- Energy is stored in organic molecules
- Heterotrophs eat organic molecules and digest them for:
  - Raw materials for synthesis
  - Fuels for energy

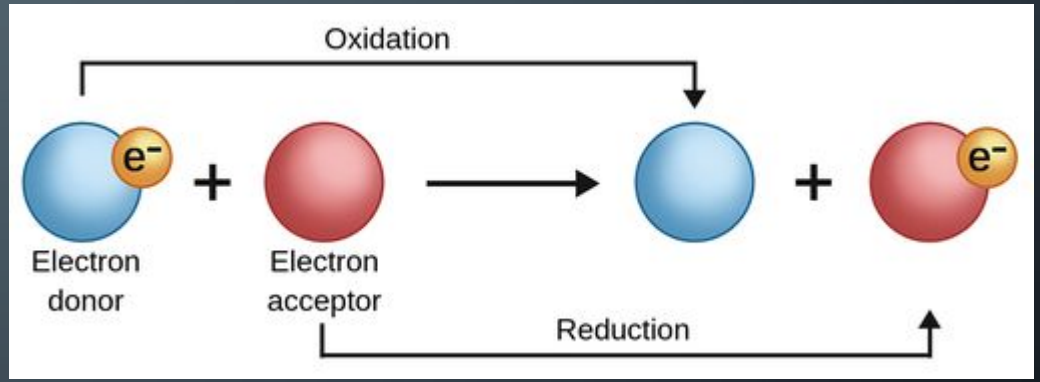
# HARVESTING STORED ENERGY

- we use catabolism of glucose as the model
- Respiration: making ATP (& some heat by burning fuels in many small steps)



# HOW DO WE HARVEST ENERGY FROM FUELS?

- Digest large molecules into smaller ones
  - Break bonds and move electrons and hydrogens from one molecule to another
  - As electrons move they carry energy  this can be stored in another bond



# OXIDATION AND REDUCTION

- Oxidation: Electrons and hydrogen are lost
- Reduction: Electrons and hydrogens are gained  energy is stored

